Q Burger: Providing premium dining experiences in smart restaurants with data analytics and AI

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- Accelerates the development process of a centralized data management platform by 3X
- Enhances restaurant turnover through instant business intelligence analytics powered by BigQuery
- Helps understand and react to customers' feedback more efficiently with Vertex AI
- Increases customer service efficiency through automatic responses with Gemini

By building a centralized data warehouse in Google Cloud to generate instant

business insights and enable AI-powered analytics, Q Burger is able to enhance its chain restaurants' operational efficiency and provide better dining experiences.

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Zoe Hsu
IT Vice General Manager, Q Burger

Since the pandemic, the demand for a frictionless digital experience in the restaurant industry has increased, with more consumers inclined to order through a restaurant's direct channels, instead of third-party apps or websites.

With that in mind, Q Burger has been dedicated to offering smooth dining experiences with the latest technologies since it launched in 2013. The company has more than 370 chain restaurants across Taiwan, and began its cloud transformation journey in 2017 to launch a series of digital services that enhance convenience for its customers, including tableside ordering and a mobile app, which had 1.1 million member users.

"We believe that digital capabilities are key to a restaurant's success and sustainability," notes Zoe Hsu, IT Vice General Manager at Q Burger. "We have implemented a number of digital strategies to build smart restaurants that provide better dining experiences. Currently, more than 60% of our orders are made online."

As the number of its chain restaurants grows, Q Burger decided in early 2024 that it would start conducting large-scale data analytics and employing artificial intelligence (AI) solutions to continue improving its operations and services through swift decision-making and process standardization. Since its business data were scattered in its on-prem databases, its point-of-sales (POS) system and several cloud platforms, Q Burger needed to build a centralized data warehouse to generate useful insights. As a result, the restaurant chain decided to implement Google Cloud for its easy integration with the SaaS products that the company was already using, its wide-ranging AI capabilities supported by high-level security and compliance, as well as the intuitive interface of the Google Cloud console.

"We always look for the best solutions in the industry to ensure that we make the most effective improvements," explains Hsu. "In terms of data analytics and AI, Google Cloud provides industry-leading tools that deliver great performance and are highly compatible with our existing systems, perfectly meeting our needs."

Maximizing turnover and efficiency with instant business intelligence analytics in BigQuery

In January 2024, Q Burger started building a centralized data warehouse aggregating all its business data in BigQuery. It used Integration Connectors to integrate all of its sources to BigQuery and completed the development of its data analytics platform in just two months. According to Hsu, this process would have taken more than six months using other on-prem or cloud tools.

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Google Cloud partner Microfusion supported the whole migration process, providing extensive technical support from the project planning stages and timely troubleshooting, to facilitate Q Burger's development and maintenance work. "Microfusion has been extremely helpful by giving constructive advice and quickly proposing solutions whenever we encounter an issue. Our data analytics platform runs so smoothly thanks to them," Hsu says.

timely to maximize their profit, our regional managers can provide guidance and assistance to individual restaurants in need more efficiently."

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Now, the revenue, raw material order and stockage data of Q Burger's chain restaurants are streamed from its POS end devices to BigQuery through Cloud Data Fusion and Cloud Run functions, while its sales and finance data are batch transferred from third-party systems through Cloud SQL to BigQuery. The data insights generated by BigQuery are then sent to Looker Studio for visualization, which is now accessible for Q Burger's executives like regional managers.

Before using Google Cloud, Q Burger used two on-prem databases to draw data from various data sources, which limited use scenarios, because not all data sources could be easily connected with its databases. With the centralized data pipelines in Google Cloud, the restaurant chain is able to run more comprehensive and instant data analytics to accelerate business decision-making. For example, its chain restaurant owners can now see their estimated monthly profit with a minor error rate every day, which allows them to take necessary measures more timely. Prior to the launch of the new data pipelines, Q Burger's restaurant owners received their income statement reports every two weeks or at the end of each month.

Q Burger's new data analytics platform has also enhanced its management efficiency. Before, its regional managers needed to review the performance of the chain restaurants that they supervised and manually single out the ones that fell under expectations. Now, by using Cloud Functions to automatically trigger alerts in Q Burger's third-party SaaS software based on analytics results produced by BigQuery, the company's regional managers can know instantly which restaurants they should pay more attention to.

"Since the launch of our new data analytics platform with BigQuery at its core, our turnover has increased," says Hsu. "Not only our chain restaurant owners are able to react more timely to maximize their profit, our regional managers can provide guidance and assistance to individual restaurants in need more efficiently."

Enabling better business expansion decisions through predictive analytics

"What's great about deploying our data warehouse in BigQuery is that we can continue broadening our data analytics range by leveraging other Google solutions, such as Google Maps Platform. We're able to launch a new type of analysis easily and gain better understanding

Leveraging the analytics capabilities of BigQuery, Q Burger has been expanding its use cases within the company. In July 2024, it retrieved the geolocation data of its chain restaurants using Places API in Google Maps Platform and sent them to BigQuery for analyses. Based on the data insights, Q Burger was able to categorize its restaurants and identify the business opportunities and risks of each category, which allows its executives to make more well-informed operational and marketing decisions.

Q Burger will also be leveraging the machine learning (ML) capabilities of BigQuery to predict business opportunities based on locations when it needs to decide where to open a new about our operations."

Zoe Hsu
IT Vice General Manager, Q Burger

restaurant. This will allow Q Burger to draw in higher returns on investment during business expansion.

"What's great about deploying our data warehouse in BigQuery is that we can continue broadening our data analytics range by leveraging other Google solutions, such as Google Maps Platform. We're able to launch a new type of analysis easily and gain better understanding about our operations," notes Hsu.

Responding to customers' feedback more quickly with Vertex AI

Q Burger's chain restaurants receive a lot of reviews every day on Google Maps. Its managers have been reading and responding to reviews manually, which takes a substantial amount of time and delays necessary actions. To react to its customers' feedback more timely, Q Burger in April 2024 began to develop a public opinion analysis system, which collects review data from Google Maps and uses Vertex AI to run sentiment analysis, identify heated topics, and generate summaries. Once the system frontend supported by Cloud Run is completed in August 2024, all the analytics results will be updated every five minutes in the system, allowing the Q Burger team to learn about its customers' dining experiences instantly and make necessary improvements.

Using Gemini 1.5 Pro, Q Burger is training a generative AI model to automatically reply to reviews on Google Maps following the brand's guidelines. This way, Q Burger is able to interact with its customers more efficiently, boosting the customer experience through automatic replies that are in line with its brand image.

Continuing operations and service optimization in Google Cloud

Leveraging Google Cloud tools, Q Burger still has several projects in the pipeline to optimize its operations and services. The restaurant chain plans to add its other business data like app member data to BigQuery to conduct more comprehensive analytics. It will also migrate the data of its POS system and mobile app to Google Cloud, so that it can build a one-stop platform where its chain restaurant owners can access all the business data they need in one place. Employing Gemini models in Vertex AI, the company will build an AI agent to answer all its employees' and restaurant owners' questions about operations, as well as a chatbot to provide instant customer services.

Hsu says, "The excellent data analytics and AI capabilities of Google Cloud has helped us significantly improve our operations and customer experiences. We believe that smart restaurants are the future, and we'll stay at the forefront of the industry by deepening our use of Google Cloud products."

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Founded in 2013, **Q Burger** has more than 370 chain restaurants in Taiwan. Dedicated to modernizing restaurant operations with the latest technologies, Q Burger launched a series of digital services that enable high convenience and efficiency for its customers, including tableside ordering and a mobile app, which has 1.1 million users.

Industry: Retail

Location: Taiwan

Products: BigQuery, Vertex AI, Cloud Data Fusion, Cloud Run functions, Cloud Run, Cloud SQL, Google Cloud console, Integration Connectors, Looker Studio

About Google Cloud Partner- Microfusion

Microfusion is a Taiwanese cloud service provider founded in 2007, offering one-stop solutions and consulting with its professional architects and maintenance team to assist customers in cloud migration, IT infrastructure modernization, and data analytics. Having worked with more than 2,000+ customers from public and private sectors, Microfusion is a service provider for digital transformation across industries.

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